Value of Money, Inflation,

Demand & Supply of Money

Meaning of Money

Money is anything generally accepted as a means of payment for goods and services and for settling debts. It plays a central role in economic activities by facilitating exchange and eliminating the problems of the barter system, especially the double coincidence of wants. In modern economies, money exists in different forms such as cash, bank deposits, and digital payments, all serving the same purpose of enabling smooth transactions.

Functions of Money

1. Medium of Exchange
Money is used to buy and sell goods and services. It removes the need for barter and simplifies transactions. This makes trade faster and more efficient.

2. Unit of Account
Money provides a common measure to express the value of goods and services. It helps in pricing, comparison, and accounting, making economic decisions easier.

3. Store of Value
Money allows individuals to save purchasing power for future use. However, its effectiveness depends on price stability, as inflation reduces its value over time.

4. Standard of Deferred Payment
Money is used to settle future payments such as loans and credit transactions. This function supports the development of financial systems and economic growth.

Meaning of Value of Money

The value of money refers to its purchasing power, meaning the quantity of goods and services that a unit of money can buy at a given time. If a certain amount of money can purchase more goods, its value is considered high. Conversely, if it can purchase fewer goods, its value is low.

For example, if 100 taka can buy 5 kg of rice today but only 3 kg next year, the value of money has decreased. This shows that the value of money is not fixed and changes over time depending on economic conditions.

Relationship Between Value of Money and Price Level

The value of money and the general price level have an inverse relationship. When the price level increases, the value of money decreases, and when the price level decreases, the value of money increases.

This relationship can be expressed as:
Value of Money $\propto$ 1 / Price Level

For example, if the price level doubles, the purchasing power of money is reduced by half. This inverse relationship explains why inflation reduces the value of money.

- **Factors Affecting the Value of Money**

1. Supply of Money
The supply of money is one of the most important determinants of its value. When the money supply in an economy increases, more money becomes available relative to goods and services. As a result, prices tend to rise, and the purchasing power of money declines.

For example, if a central bank prints excessive money without a corresponding increase in production, inflation occurs. This situation is often described as “too much money chasing too few goods.” On the other hand, a controlled money supply helps maintain price stability and preserves the value of money.

2. Demand for Money
The demand for money refers to the desire of individuals and businesses to hold money for transactions, precautionary needs, and speculative purposes. When the demand for money increases, people prefer to hold cash rather than spend it.

As a result, spending decreases, which puts downward pressure on prices and increases the value of money. Conversely, when the demand for money falls, people spend more, increasing prices and reducing the value of money.

3. Price Level
The general price level directly determines the value of money. There is an inverse relationship between price level and purchasing power.

When prices rise (inflation), each unit of money buys fewer goods and services, reducing its value. When prices fall (deflation), the same amount of money can buy more goods, increasing its value.

4. Volume of Trade and Output
The level of production and trade in an economy also affects the value of money. When the supply of goods and services increases due to higher production, prices tend to stabilize or even fall, which increases the value of money.

In contrast, low production or supply shortages can lead to higher prices and a decline in the value of money.

5. Velocity of Circulation
Velocity of circulation refers to the speed at which money changes hands in an economy. When money circulates rapidly, it increases the effective supply of money in the economy.

A higher velocity means more spending, which can lead to higher prices and a reduction in the value of money. On the other hand, when money circulates slowly, spending decreases, which can stabilize prices and increase the value of money.

6. Government Policies

Government policies, particularly monetary and fiscal policies, play a significant role in determining the value of money.

Monetary policy includes interest rates, open market operations, and reserve requirements. Fiscal policy includes government spending and taxation. Effective policy management is essential to maintain the stability and value of money.

- **Fisher's quantity theory of money:**

Fisher's Quantity Theory of Money is one of the most important theories in monetary economics. It explains the relationship between the money supply and the general price level. The theory was developed by economist Irving Fisher and provides a simple framework to understand inflation.

The Fisher Equation

MV = PT

Where:
M = Money Supply (total money in circulation)
V = Velocity of Money (number of times money is used)
P = Price Level (average prices of goods and services)
T = Volume of Transactions (total goods and services exchanged)

****Explanation of the Theory**

The equation shows that total money spent in the economy (MV) is equal to the total value of goods and services (PT).

If the money supply increases while velocity and transactions remain constant, the price level must increase, leading to inflation.

For example, if money supply increases by 10% and other factors remain constant, the price level will also increase by approximately 10%. This explains the idea of 'too much money chasing too few goods'.

*** Assumptions of the Theory**

- Constant Velocity: Velocity of money is stable.
- Fixed Transactions: Output is constant in the short run.
- Full Employment: Economy operates at full capacity.
- No Hoarding: All money is actively used.

****Criticisms of the Theory**

- Unrealistic Assumptions: Full employment and constant output rarely exist.
- Ignores Demand for Money: Does not consider why people hold money.

- Velocity Is Not Constant: It changes over time.
- Oversimplified Model: Modern economies are more complex.

Role of Money in Economic Development

Money plays a crucial role in promoting economic development. It is not only a medium of exchange but also a key driver of production, investment, and overall economic growth. A well-functioning monetary system helps allocate resources efficiently and supports long-term development.

Key Roles of Money

1. Promotes Trade and Exchange
Money eliminates the limitations of the barter system and enables smooth exchange of goods and services. It allows both domestic and international trade to expand, increasing economic activity.

Example: A farmer can sell crops for money and use it to buy machinery or services easily.

2. Facilitates Investment
Money helps channel savings into productive investments. Financial institutions collect savings and provide loans to businesses, leading to capital formation and economic growth.

Example: Banks provide loans to industries to expand production and create employment.

3. Encourages Savings
Money acts as a store of value, encouraging individuals to save for future needs. Savings are essential for investment and long-term economic development.

Example: People deposit money in banks, which is later used for business financing.

4. Supports Financial Markets
Money is the foundation of financial systems such as banks, stock markets, and bond markets. These institutions mobilize funds and allocate them efficiently across the economy.

Example: Stock markets help companies raise capital for expansion.

5. Enhances Specialization and Productivity
Money enables specialization by allowing individuals to focus on specific activities and exchange their output for money. This increases productivity and efficiency.

Example: Workers specialize in different sectors and use money to meet their diverse needs.

6. Helps in Economic Planning
Money provides a standard measure for economic calculation and planning. Governments and businesses use monetary values to make decisions regarding production, investment, and policy.

**Definition of Inflation

Inflation refers to a continuous and sustained rise in the general price level of goods and services over a period of time. It does not mean a one-time increase in the price of a single good; rather, it reflects a broad increase in prices across the entire economy.

As inflation rises, the purchasing power of money declines, meaning consumers can buy fewer goods and services with the same amount of money. For example, if inflation increases by 10%, a product that previously cost 100 taka will now cost 110 taka.

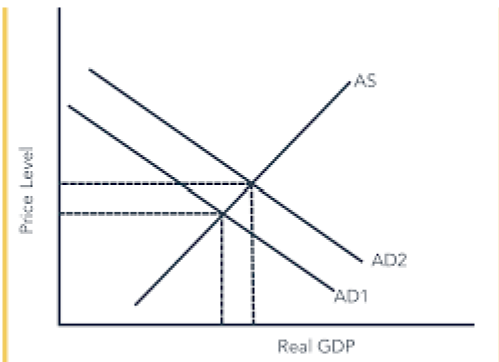
Inflation is measured through price indices such as the Consumer Price Index (CPI) and the GDP Deflator.

** Types of Inflation

1. Demand-Pull Inflation

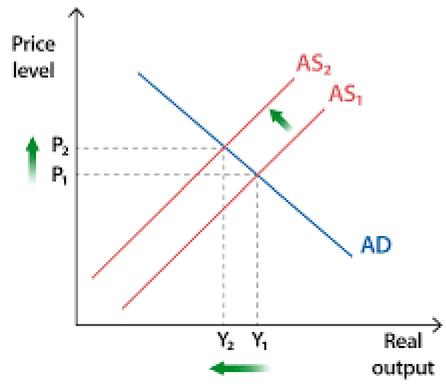
Demand-pull inflation occurs when aggregate demand exceeds aggregate supply in the economy. When consumers, businesses, and the government increase their spending, demand rises faster than production, leading to higher prices.

This situation is often described as 'too much money chasing too few goods'.



2. Cost-Push Inflation

Cost-push inflation occurs when the cost of production increases, forcing firms to raise prices. This may be due to higher wages, rising fuel prices, or increased taxes.



Firms pass these higher costs on to consumers, resulting in inflation.

3. Inflation Based on Degree
Creeping Inflation: Low and stable inflation.
Walking Inflation: Moderate level of inflation.
Running Inflation: High and harmful inflation.
Hyperinflation: Extremely high and uncontrollable inflation.

****Causes of Inflation**

- Increase in Money Supply: Excess money in the economy increases demand and raises prices.
- High Aggregate Demand: Increased consumption and government spending push prices upward.
- Rising Production Costs: Higher wages and input costs increase prices.
- Supply Shocks: Disruptions such as natural disasters reduce supply and increase prices.

****Impact of Inflation**

Positive Effects
Encourages production and investment.
Reduces real burden of debt.
Stimulates spending.

Negative Effects
Reduces purchasing power.
Increases inequality.
Creates uncertainty.
Reduces value of savings.
Hurts export competitiveness.

****Control Measures of Inflation**

- Monetary Policy: Increase interest rates and reduce money supply.
- Fiscal Policy: Reduce government spending and increase taxes.
- Price Control: Government fixes prices temporarily.
- Increase Production: Boost supply of goods and services.

Cost of Inflation

1. Shoe-Leather Cost
Refers to the inconvenience and cost of reducing money holdings during inflation. People frequently withdraw money or shift funds, increasing time and transaction costs.

2. Menu Cost
Costs incurred by firms when changing prices frequently. Includes updating menus, catalogs, price tags, and billing systems.

3. Loss of Purchasing Power
Inflation reduces real income. Fixed-income earners and low-income groups suffer most as their money buys fewer goods.

4. Uncertainty and Economic Instability
High inflation creates uncertainty. It discourages investment, savings, and long-term planning.

5. Income Redistribution
Inflation benefits debtors and harms creditors. It increases inequality between different income groups.

6. Distortion of Price Signals
Inflation makes it difficult to distinguish real price changes, leading to poor economic decisions.

7. Impact on Savings and Investment
Inflation discourages saving and reduces long-term investment, affecting economic growth.

****Definition of Deflation**
Deflation refers to a sustained and continuous fall in the general price level of goods and services over time. Unlike a temporary decline in the price of a few goods, deflation reflects a broad-based decrease in prices across the economy.

Although falling prices may appear beneficial, deflation is often associated with weak aggregate demand, declining income, and economic slowdown. As a result, it can have serious negative consequences for economic stability.

****Characteristics of Deflation**
Deflation is characterized by several important features:
- Continuous decline in general price level
- Increase in purchasing power of money
- Reduction in aggregate demand
- Decline in income and profits

- Rising unemployment
- Weak economic activity

*****Causes of Deflation**

1. Insufficient Aggregate Demand
When consumers and businesses reduce their spending, overall demand in the economy declines. This forces producers to lower prices in order to sell their goods and services.

2. Excess Supply of Goods
When production exceeds demand, firms face unsold inventories. To clear excess stock, they reduce prices, leading to deflation.

3. Credit Contraction
When banks reduce lending or tighten credit conditions, less money is available for spending and investment. This reduces demand and contributes to falling prices.

4. Decline in Investment
A fall in business investment leads to lower production, income, and employment. This further reduces demand and reinforces deflationary pressures.

****Effects of Deflation**

1. Increase in Purchasing Power
As prices fall, the purchasing power of money increases. However, this benefit is often offset by declining income and employment.

2. Decline in Production
Firms reduce output due to falling demand and declining profits, leading to lower economic activity.

3. Rise in Unemployment
Reduced production forces firms to cut labor costs, resulting in higher unemployment.

4. Increase in Real Debt Burden
Deflation increases the real value of money, making it more difficult for borrowers to repay loans. This puts financial stress on households and businesses.

5. Economic Slowdown
Deflation can create a vicious cycle of falling demand, production, income, and employment, leading to prolonged economic stagnation.

****Control Measures of Deflation**

1. Increase Government Spending
Expansionary fiscal policy increases demand by injecting money into the economy through public spending.

2. Reduce Interest Rates
Lower interest rates encourage borrowing, investment, and consumption, helping to boost demand.

3. Increase Money Supply
Central banks may increase the money supply through policies such as quantitative easing to stimulate economic activity.

4. Encourage Investment
Providing incentives such as tax benefits and subsidies can encourage businesses to invest and expand production.